Whose Side Is the Bidding Assistant On?

Sponsor objectives, advice following and equilibrium selection in a first-price auction

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GitHub: github.com/dku-comsci-econ206-Autumn2026/ps2-ai-advice-auction Demo: huggingface.co/spaces/dku-comsci-econ206-2026/ps2-ai-advice-auction

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Two first-price pure equilibria. Sponsor advice changes modeled distribution while welfare stays 16. Human behavior and live-LLM effects remain untested.

01 Question, model and contribution

Research question and strategic setting

How do disclosed buyer/seller objectives, advice following and beliefs interact with auction rules to redistribute surplus?

Two buyers. Value 6, budget 5, bids 2 or 5, in 10 RMB units. Highest bid wins and pays its bid. Ties split 50/50. One round. Values and common disclosed advice are common knowledge.

Buyer 1 / Buyer 2	Low 2	High 5
Low 2	7, 7 Nash	5, 6
High 5	6, 5	5.5, 5.5 Nash

High is strictly better if P(other bids High) > 2/3.

At equality either bid is a best reply.

Three lenses and auction comparison

Game theory: two pure equilibria and a mixed boundary.
Social choice: buyer payoff versus seller revenue, welfare 16.
Mechanism design: second price makes High strictly dominant within the restricted two-bid menu.

Intellectual Merits diagram

INHERITED FOUNDATION
Kuang: interested advice
Chaudhuri: common knowledge
Malenko–Tsoy: advised buyers

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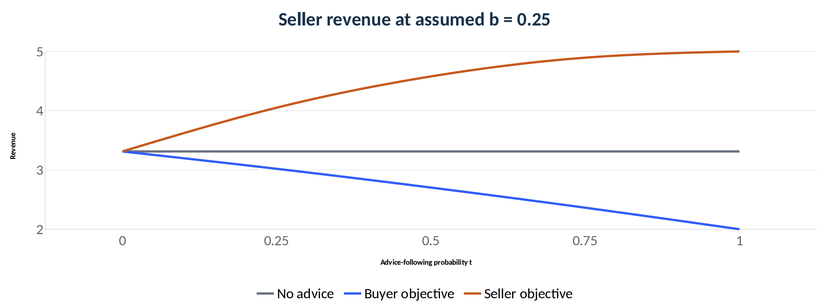
FP2 ADDITION
Sponsor-disclosed coordination auction. Compare rules and buyer/seller distribution.

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EVIDENCE
Matrix, deviation checks, p* = 2/3, t*(b), exact expectations. No human or AI-specific effect established.

02 Computation and behavioral instrument

Exact model expectations



Exact expectations at the shown t values. Curves meet at t = 0, revenue 3.3125. The 5/9 value is a best-reply threshold, not a revenue crossing. Intermediate independent following need not be equilibrium play.

New checked baseline sensitivity

Assumed baseline b	0	0.25	0.5
Threshold t*	2/3	5/9	1/3

Exact calculation. Twelve automated tests pass. At full following, buyer advice gives payoff 7 and revenue 2; seller advice gives payoff 5.5 and revenue 5. Total welfare stays 16.

Revised Space protocol and evidence boundary

Before revealing outcomes: condition, belief about the other buyer, trust rating, bid and explanation. Optional no-advice-first order. Manual pairing and explicit CSV export. No participant dataset collected. The deployed Space implements this protocol.

Closest papers and source links

Kuang, X., Weber, R.A., Dana, J. (2007). How effective is advice from interested parties? JEBO 62(4), 591–604. doi:10.1016/j.jebo.2005.03.010
Chaudhuri, A., Schotter, A., Sopher, B. (2009). Talking ourselves to efficiency. Economic Journal 119(534), 91–122. doi:10.1111/j.1468-0297.2008.02207.x
Malenko, A., Tsoy, A. (2019). Selling to advised buyers. AER 109(4), 1323–1348. doi:10.1257/aer.20170334

03 Impact, roadmap and dialogue

Practical Impacts diagram

PROPOSED RULE
Disclose sponsor. Offer both objectives. Keep inspectable recommendation records.

16 PEACE, JUSTICE AND STRONG INSTITUTIONS

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STAKEHOLDERS AND INDICATORS
Buyers, sellers and platform. Sponsor recognition, record completeness and payoff gaps.

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SDG 16, TARGET 16.6
Proposed accountable and transparent advice. Real-world benefits remain unmeasured.

Baseline: model buyer payoff 6.34375, revenue 3.3125 at assumed b = 0.25. Participant baseline unknown.
Project indicators differ from official UN indicators.
Risks: unwarranted reliance, seller-favoring transfers, identifying logs. Safeguards: opt-out, both advice views, minimal records and a correction route.

New Future Research Roadmap diagram

LIMITATION
Trust, opponent beliefs and advice-source labels can produce similar bids.

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ECONOMICS
Vary budget/value/menu.
Output: equilibrium sensitivity table.

CS
Log fixed-prompt LLM advice.
Output: deviations and uncertainty.

BEHAVIOR
Elicit beliefs. Match advice, vary source.
Output: reviewed pilot.

Milestones: freeze assumptions and protocol; publish a checked benchmark; run an exploratory pilot. Possible collaborators: Han Zhang, Yongzhuo Wu or disciplinary researchers, if interested. No collaboration commitment is claimed.

Feedback, recognition and limitations

Yongzhuo Wu / FP1 awarded the Model Clarity Award on Oct 12 and offered pilot-design review. Keep verified equilibria, revise novelty/figures, investigate beliefs and source effects. Common value, two bids and independent following limit inference. Exact records and missing classroom evidence: paper Appendix G.

Named contributions

Yitong Lin defined the auction and advice mechanism, developed the GitHub and Hugging Face artifacts and main paper, and checked computational consistency. His reported work connects the strategic model to computation and the proposed behavioral study.

Qingyue Liu abstracted the setting into a two-player game, designed the payoff matrix and initial outline, revised the paper and designed the poster. She checked equilibria, condition labels and interpretation, providing the shared model and visual structure.

Full acknowledgments

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